Final Project Report:
How Music Taste Changes Over Time

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Abstract

Music is a key part of daily life, yet people's opinions about music have evolved over time. With factors, such as race, having a different impact on society during each decade, this study focuses on how various factors affect song ratings during different decades. Data from Billboard Hot 100 Number One songs were taken from the 1960s - 2010s and split up to form rating prediction linear models. We compared main effects models with interaction effects models and decided which one would be more viable to optimize. The chosen model for the decade was optimized based on its AIC score to determine which predictors made a significant impact on the model. Comparing the predictors across models revealed that variables such as energy and race were consistently important across most decades. However, no singular factor was consistent between every decade. Ultimately, the findings demonstrated that while the factors that predict song ratings change over time, they have the potential to be similar in different time periods.

1 Introduction

Research Questions:

1. What are the most and least optimal predictors over time from 1960-2010?
2. How do the models stay consistent and how do they change?
3. How does the predictive power of the predictors change over time and what model has the best?

Our interest in music inspired our group to investgate its distinction over time. This prompted dividing our data into different decades to group similar song eras together, allowing us to understand what characteristics truly impact the overall rating. Provided that music is vastly different in each generation, it is interesting to see what aspects of music and the artist allow us to predict how a song is rated. This study shows the historical differences between predictors and how aspects, such as society and race can affect how a song is rated.

2 Data Description and Exploratory Data Analysis (EDA)

Data Description:

This dataset includes information about every Number 1 song on the Billboard charts from 1958-2025. It was compiled by Chris Dalla Riva when he decided to listen to each number one hit song and track data related to each song in a dataset. Most of the data comes from facts about the song, Chris's interpretation about the song, or spotify data and ratings. The data helps with determining what factors lead to a specific song being on top of the billboards. It accounts demographic data such as age, gender, and race, as well as looking at specific song statistics, such as energy.

2.1 Variables

Let Y denote the overall rating of the song. The primary features are:

- Age: Average age of the artist(s)
- Gender: Gender of the artist(s)
- Race: Race of the artist(s)
- Genre: Type of genre of the song
- Energy: Energy measure from Spotify (0-100)
- Danceability: Danceability measure from Spotify (0-100)
- Happiness: Happiness measure from Spotify (0-100)
- Loudness_db: Loudness measured in decibels from Spotify
- BPM: Beats per minute from Spotify
- Length_sec: Length of the song in seconds

However, in our models, the predictors are split into:

- X_1: Age: Average age of the artist(s)
- X_2: GenderNotMale: Whether any of the artists aren't male (1 if not male, 0 otherwise)
- X_3: RaceWhite: Whether any of the artists are white (1 if white, 0 otherwise)
- X_4: GenreJazz: Whether the song is jazz (1 if jazz, 0 otherwise)
- X_5: GenreRock: Whether the song is rock (1 if rock, 0 otherwise)
- X_6: GenrePop: Whether the song is pop (1 if pop, 0 otherwise)
- X_7: GenreOther: Whether the song is a different genre (1 if not rock, pop, jazz or country and 0 otherwise)
- X_8: Energy: Energy measure from Spotify (0-100)
- X_9: Danceability: Danceability measure from Spotify (0-100)
- X_{10}: Happiness: Happiness measure from Spotify (0-100)
- X_{11}: Loudness_db: Loudness measured in decibels from Spotify
- X_{12}: BPM: Beats per minute from Spotify
- X_{13}: Length_sec: Length of the song in seconds

2.2 Exploratory Analysis

Present initial findings from your EDA. You can reference figures like scatterplots or correlation matrices here (e.g., "As seen in Figure **??**...").

3 Methodology and Model Building

Model Fitting Procedure

1. Fit a main effects and interaction effects model to predict the decade's overall song ratings using the variables states above
2. Determine whether the interaction effects model included substatial predictors that weren't included in our main effects model by running an anova test and interpretting the p-value (at a signficance level of 0.1)

3. If the p-value was significant, then run a backwards step procedure on the interaction effects model to remove insignificant predictors based on the change in AIC per step. Otherwise, run the backwards step procedure on the main effects model.

4. Repeat the previous steps for each decade to acquire the optimized models

1960s Model

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \beta_5 X_{i5} \\ &+ \beta_6 X_{i6} + \beta_7 X_{i7} + \beta_8 X_{i8} + \beta_9 X_{i9} + \beta_{10} X_{i10} + \beta_{11} X_{i13} \\ &+ \beta_{12}(X_{i3}X_{i4}) + \beta_{13}(X_{i3}X_{i7}) + \beta_{14}(X_{i3}X_{i6}) \\ &+ \beta_{15}(X_{i3}X_{i8}) + \beta_{16}(X_{i3}X_{i9}) + \beta_{17}(X_{i2}X_{i9}) \\ &+ \beta_{18}(X_{i2}X_{i10}) + \epsilon_i \end{aligned} \tag{1}$$

1970s Model

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 X_{i1} + \beta_2 X_{i3} + \beta_3 X_{i4} + \beta_4 X_{i5} \\ &+ \beta_5 X_{i6} + \beta_6 X_{i7} + \beta_7 X_{i13} + \epsilon_i \end{aligned} \tag{2}$$

1980s Model

$$Y_i = \beta_0 + \beta_1 X_{i8} + \beta_2 X_{i9} + \beta_3 X_{i10} + \epsilon_i \tag{3}$$

1990s Model

$$Y_i = \beta_0 + \beta_1 X_{i2} + \beta_2 X_{i3} + \beta_3 X_{i8} + \beta_4 X_{i10} + \beta_5 X_{i11} + \epsilon_i \tag{4}$$

2000s Model

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i5} \\ &+ \beta_5 X_{i6} + \beta_6 X_{i7} + \beta_7 X_{i8} + \beta_8 X_{i9} + \beta_9 X_{i10} + \beta_{10} X_{i12} \\ &+ \beta_{11} X_{i13} + \beta_{12}(X_{i1}X_{i9}) + \beta_{13}(X_{i3}X_{i7}) \\ &+ \beta_{14}(X_{i3}X_{i9}) + \beta_{15}(X_{i3}X_{i10}) + \beta_{16}(X_{i3}X_{i12}) \\ &+ \beta_{17}(X_{i3}X_{i13}) + \beta_{18}(X_{i2}X_{i9}) + \beta_{19}(X_{i2}X_{i10}) \\ &+ \beta_{20}(X_{i2}X_{i12}) + \epsilon_i \end{aligned} \tag{5}$$

2010s Model

$$Y_i = \beta_0 + \beta_1 X_{i2} + \beta_2 X_{i8} + \beta_3 X_{i10} + \beta_4 X_{i13} + \epsilon_i \tag{6}$$

3.1 Proposed Model

We chose the model that best explained the variability in overall ratings in its decade while still being penalized on complexity (best adjusted R^2 value). According to the adjusted r^2 values, the linear model predicting overall song ratings in the 60s was the best model.

Therefore, we fit a multiple linear regression model of the form:

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \beta_5 X_{i5} \\ &+ \beta_6 X_{i6} + \beta_7 X_{i7} + \beta_8 X_{i8} + \beta_9 X_{i9} + \beta_{10} X_{i10} + \beta_{11} X_{i13} \\ &+ \beta_{12}(X_{i3}X_{i4}) + \beta_{13}(X_{i3}X_{i7}) + \beta_{14}(X_{i3}X_{i6}) \\ &+ \beta_{15}(X_{i3}X_{i8}) + \beta_{16}(X_{i3}X_{i9}) + \beta_{17}(X_{i2}X_{i9}) \\ &+ \beta_{18}(X_{i2}X_{i10}) + \epsilon_i \end{aligned} \tag{7}$$

where $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Explain how you selected your variables (e.g., stepwise selection, theoretical justification, handling of multicollinearity). If you used transformations (like taking the log of the response), explain why.

4 Results and Diagnostics

Since the 1960s model had the highest adjusted R^2 value, we will use present the results of that model and use as our final model.

4.1 Model Summary

```
 =====================================
  SUMMARY FOR OPTIMAL MODEL: 1960s
=====================================

Call:
lm(formula = rating ~ age + gender + race + genre + energy +
    danceability + happiness + length_sec + race:genre + race:energy +
    race:danceability + gender:danceability + gender:happiness,
    data = subpop)

Residuals:
    Min      1Q  Median      3Q     Max
-5.3064 -1.2960  0.1313  1.2436  4.2045

Coefficients: (1 not defined because of singularities)
                             Estimate Std. Error t value Pr(>|t|)
(Intercept)                  6.235892   2.001360   3.116  0.00214 **
age                         -0.036813   0.021122  -1.743  0.08309 .
genderNot Male              -1.820687   1.293523  -1.408  0.16101
raceWhite                    1.604179   1.795663   0.893  0.37287
genreJazz                    1.473039   1.332354   1.106  0.27040
genreOther                   1.978211   0.999147   1.980  0.04926 *
genrePop                     0.988531   0.889417   1.111  0.26788
genreRock                    0.260014   0.629236   0.413  0.67994
energy                      -0.005765   0.015336  -0.376  0.70743
danceability                -0.013030   0.022465  -0.580  0.56264
happiness                    0.004845   0.009013   0.538  0.59154
length_sec                   0.008965   0.003791   2.365  0.01912 *
raceWhite:genreJazz         -1.353373   2.222521  -0.609  0.54334
raceWhite:genreOther        -3.249204   1.187123  -2.737  0.00683 **
raceWhite:genrePop          -1.786329   0.784348  -2.277  0.02395 *
raceWhite:genreRock                NA         NA      NA       NA
raceWhite:energy             0.024399   0.017347   1.407  0.16132
raceWhite:danceability      -0.046982   0.022464  -2.091  0.03791 *
genderNot Male:danceability  0.076584   0.023552   3.252  0.00137 **
genderNot Male:happiness    -0.030694   0.016106  -1.906  0.05830 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.797 on 178 degrees of freedom
Multiple R-squared:  0.314,     Adjusted R-squared:  0.2446
F-statistic: 4.527 on 18 and 178 DF,  p-value: 4.551e-08
```

Figure 1: Summary of 1960's Optimized Model

4.2 Model Diagnostics

The figures below are the TA and QQ-plot for our selected optimized model from the 1960s.

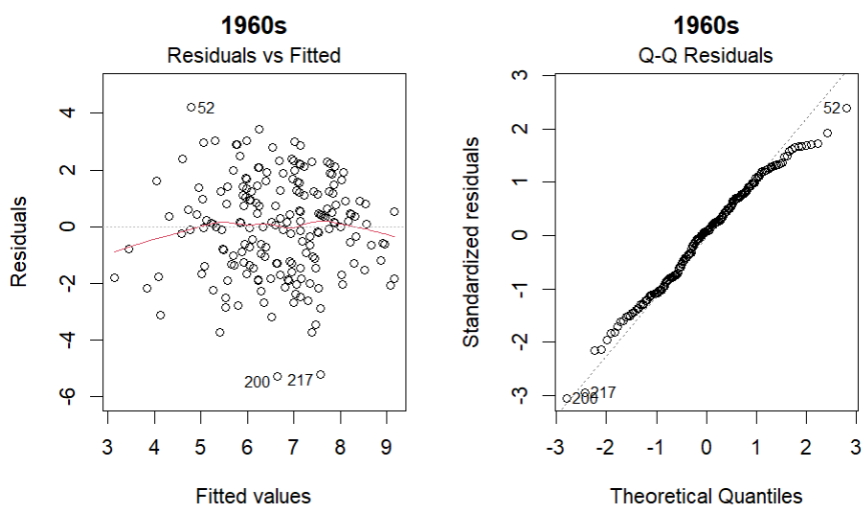

Figure 2: 1960's TA and QQ-Plot

In the TA plot, we see that the residuals are mostly around 0, which justifies the zero mean assumption. The TA plot also shows that our observations do not follow a systematic pattern and there are not many notable outliers, justifying homoscedasticity. With the QQ plot, we noticed that the tails are somewhat short-tailed on both ends for our final model. Because of this, we violate the normality assumption. However, since the short tails are without skew on the QQ plot, we believed that transformations were unnecessary for the final model.

5 Discussion and Conclusion

Summarize the main takeaways from your model in the context of the original problem. What do the coefficients mean in the real world?

Acknowledge the limitations of your study (e.g., potential omitted variable bias, non-representative sample) and suggest areas for future work.

References

Chris Dalla Riva "Billboard Hot 100 Number Ones" *https://github.com/rfordatascience/tidytuesday/tree/main/data/2025/2025-08-26*, 2025.

A Appendix: R/Python Code

Paste your analysis code here. If you prefer, you can use the `verbatim` environment to format it as raw text:

```
# Example R Code
model <- lm(Y ~ X1 + X2, data = mydata)
summary(model)
plot(model)
```

A.1 Cleaning Data

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
    orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                      ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                      ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                      ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                      ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
                danceability = orig_data$danceability, happiness = orig_data$happiness,
                loudness_db = orig_data$loudness_d_b,
                length_sec = orig_data$length_sec)


cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
```

```
                    unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]
```

A.2 Finding Optimized Models

```
model_results <- list()


for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- cleaned_data[cleaned_data$year >= i - 10 &
                         cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
  if(nrow(subpop) < 50) {
    message(paste("Skipping", decade_label, "- Not enough data points."))
    next
  }

  factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
                 function(x) length(unique(na.omit(x))))

  if(any(factor_counts < 2)) {
    message(paste("Skipping", decade_label, "due to single-level factor:",
                 names(factor_counts)[factor_counts < 2]))
    next
  }

  first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
                 + danceability + happiness + bpm + loudness_db + length_sec)

  second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
                + age*energy + age*danceability + age*happiness + age*bpm
```

```
                 + age*loudness_db + age*length_sec + race*genre + race*energy
                 + race*danceability + race*happiness + race*bpm + race*loudness_db
                 + race*length_sec + gender*genre + gender*energy + gender*danceability
                 + gender*happiness + gender*bpm +gender*loudness_db + gender*length_sec)


  if(anova(first_model, second_model)$"Pr(>F)"[2] <= 0.1){
     optimal_model <<- step(second_model, direction="backward")
  } else {
      optimal_model <<- step(first_model, direction="backward")
  }


  model_results[[decade_label]] <- list(
    "first"   = first_model,
    "second"  = second_model,
    "optimal" = optimal_model
  )

  print(decade_label)
  summary(first_model)
  summary(second_model)
  summary(optimal_model)
}
```

A.3 Response Analysis

```
ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 1,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
```

```
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
\end{verbatime

\subsection{Adjusted $R^2$ Comparison}
\begin{verbatim}
summary(model_60s)$adj.r.squared

summary(model_70s)$adj.r.squared

summary(model_80s)$adj.r.squared

summary(model_90s)$adj.r.squared

summary(model_2000s)$adj.r.squared

summary(model_2010s)$adj.r.squared

library(ggplot2)

df <- data.frame(
  Decade = factor(c("1960s","1970s","1980s","1990s","2000s","2010s"),
                  levels = c("2010s","2000s","1990s","1980s","1970s","1960s")),
  Adj_R2 = c(0.244647, 0.2013403, 0.009698796, 0.2152039, 0.2197517, 0.06362834)
)

ggplot(df, aes(x = Decade, y = Adj_R2)) +
  geom_col(fill = "#23DC6D") +
  geom_text(aes(label = round(Adj_R2,3)), hjust = -0.1, size = 4) +
  coord_flip() +
  ylim(0, 0.3) +
  labs(
    title = "Adjusted R2 of Song Rating Models by Decade",
```

```
  x = "Decade",
  y = "Adjusted R2"
) +
theme_minimal(base_size = 14)
```